



Intelligent Solutions for N108 Bank

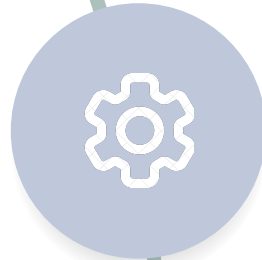
Group 6: Ariel Xiao, Catherine Yuan, Norah Zhou,
Parris McKenley, Yixuan Zhang

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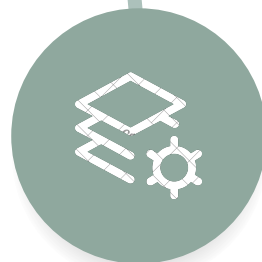
Intelligent Solutions
for N108 Bank



01 **Context & Scenarios**



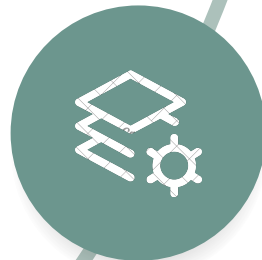
02 **Information System Architecture**



03 **Application System Architecture**



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01

Context & Scenarios

1.1 Overview



Founded **1994** as a credit card company
Completed full cloud migration to AWS in **2020**



Strategic identity:
A technology and data company that happens to run a bank

\$39.1B

Total Revenue
in 2024

~600

Cross-functional
agile team



Three core business segments:
Credit Card, Consumer Banking, Commercial Banking

1.2 Current Challenges

Integration gaps

- The past migration creates data silos, mismatched standards, and inconsistent workflows across business units

Decision Bottlenecks

- Manual intervention in credit, risk, and customer operations limits the bank's ability to meet real-time digital service demands.

Data governance gaps & Cybersecurity risk

- 2019 breach: 100M+ customers exposed via misconfigured firewall
- Ongoing need to bridge gaps between technical infrastructure and governance oversight to prevent data breaches.

Scenario 1: Post-Merger Integration

Proposed Solution:

Agent-Assisted Integration Platform

Scenario 2: Digital Transformation

Proposed Solution:

Agentic Systems

1.3 Scenario 1

Scenario 1: Post-merger integration

External trigger

Industry consolidation and mergers

Internal factors

Need to expand services and access more data



Challenge

- Legacy systems and data inconsistency
- Integration and coordination difficulties

Impact on business and organization

- More complex services and customer journeys
- Stronger need for cross-unit coordination

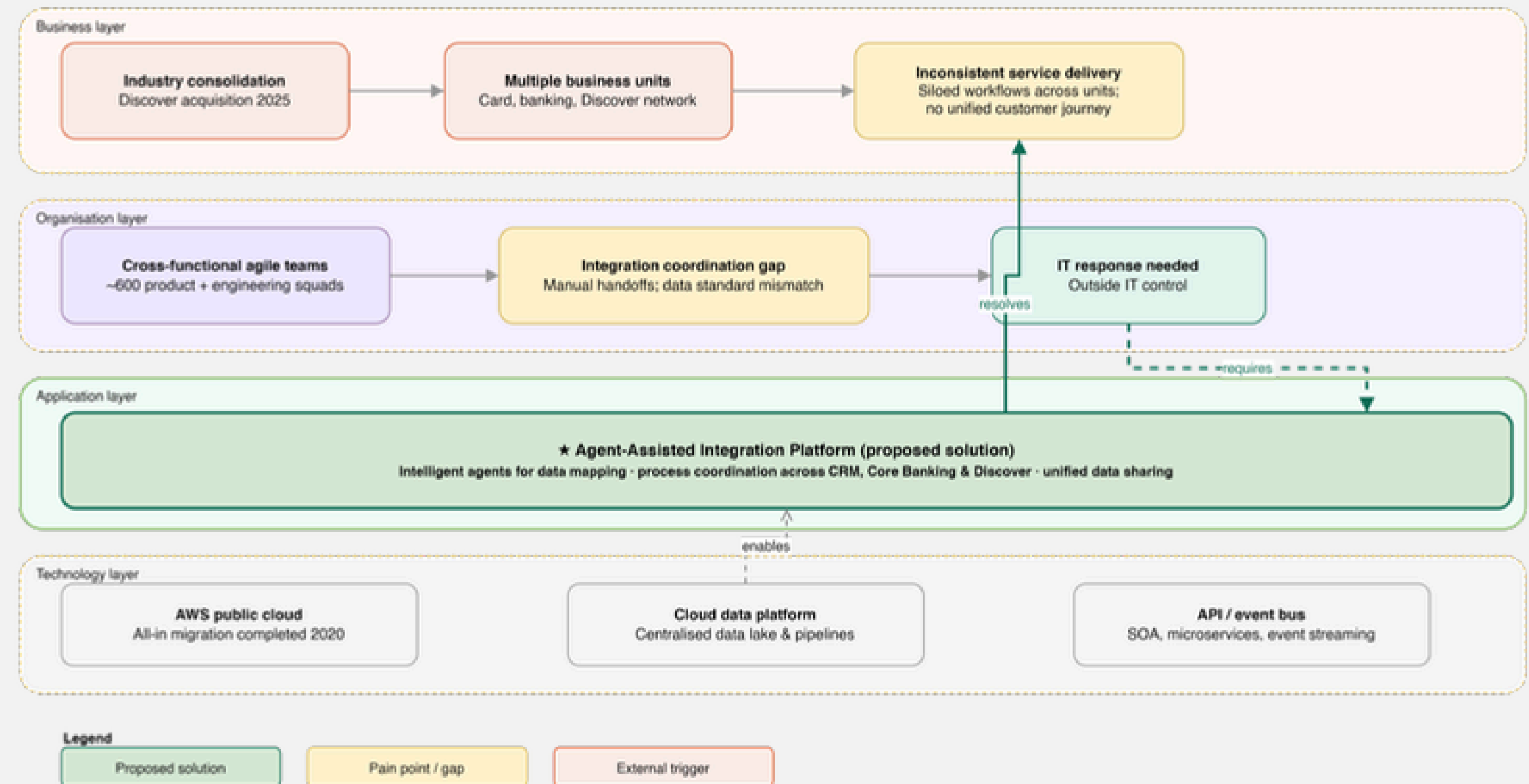
(Lankhorst et al., 2018)
(Murphy, 2014)

1.4 Solution 1:

Solution 1: Agent-assisted Integration Platform

Supports system integration and data sharing

Improves workflow coordination across systems



1.5 Scenario 2

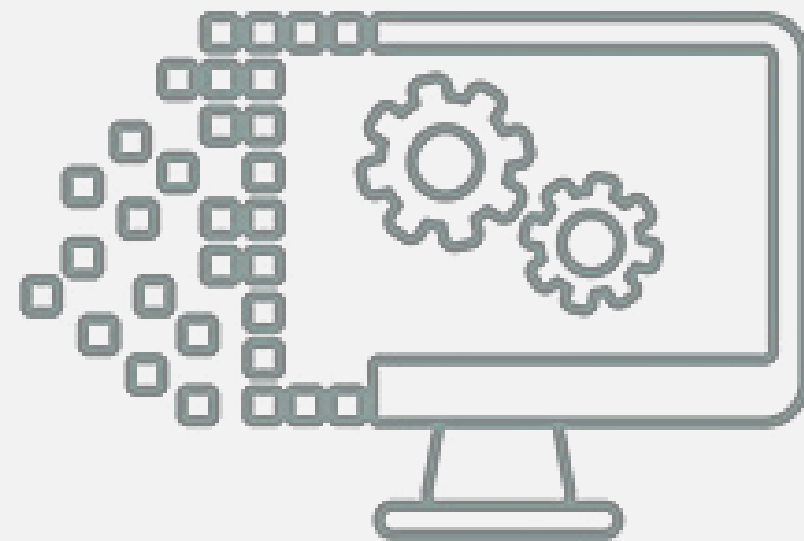
Scenario 2: Digital transformation and flatter structure

External trigger

Demand for real-time and personalized services

Internal factors

Agile practices and efficiency goals



Challenge

- Human decision-making is too slow
- Cannot meet real-time requirements

Impact on business and organization

- More decentralized decision-making
- Increased collaboration across teams

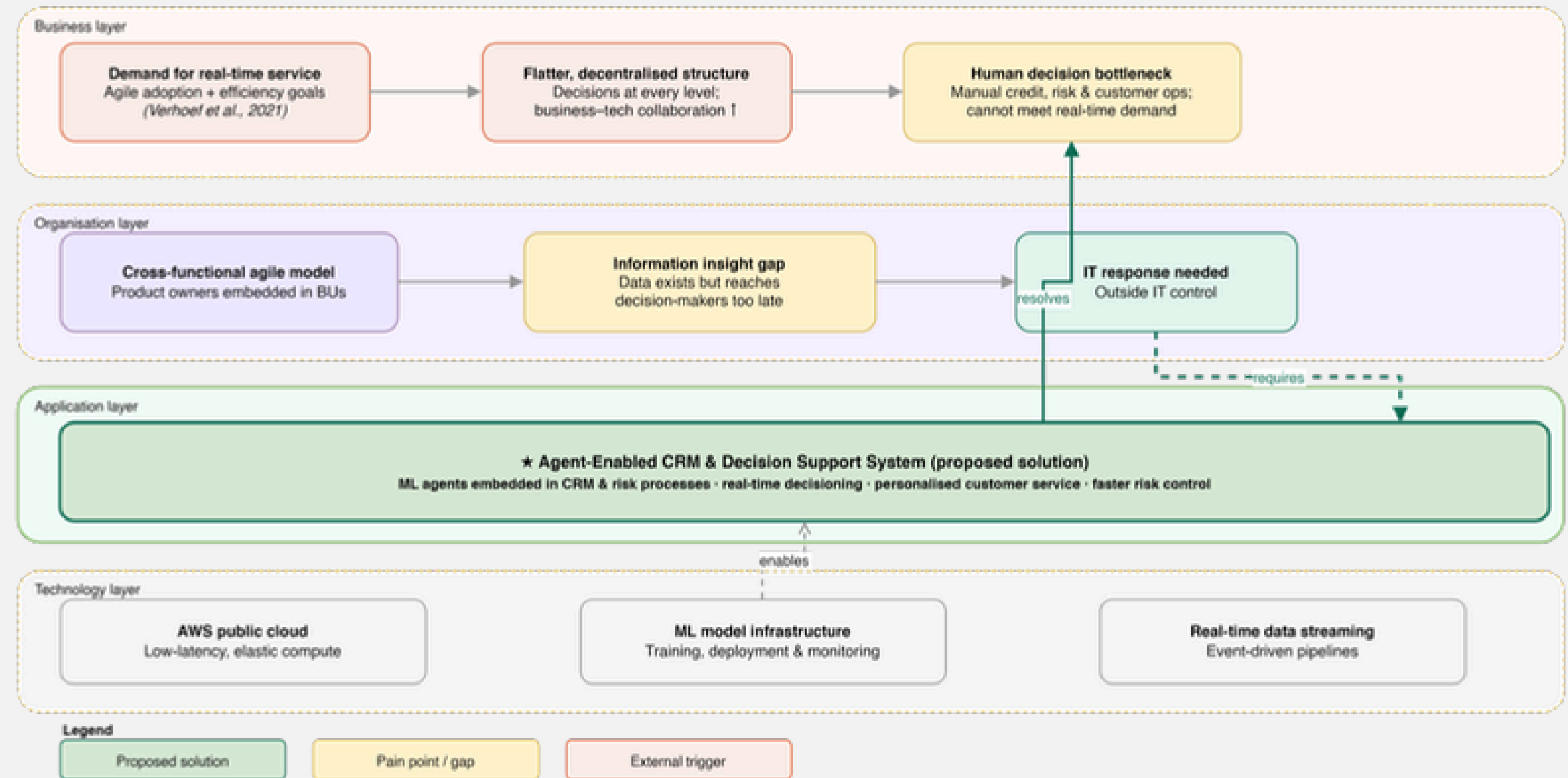
(Verhoef et al., 2021)
(Westerman et al., 2014)

1.6 Solution 2:

Solution 2: Agent-enabled CRM & Decision Support System

Supports real-time analysis

Supports real-time analysis

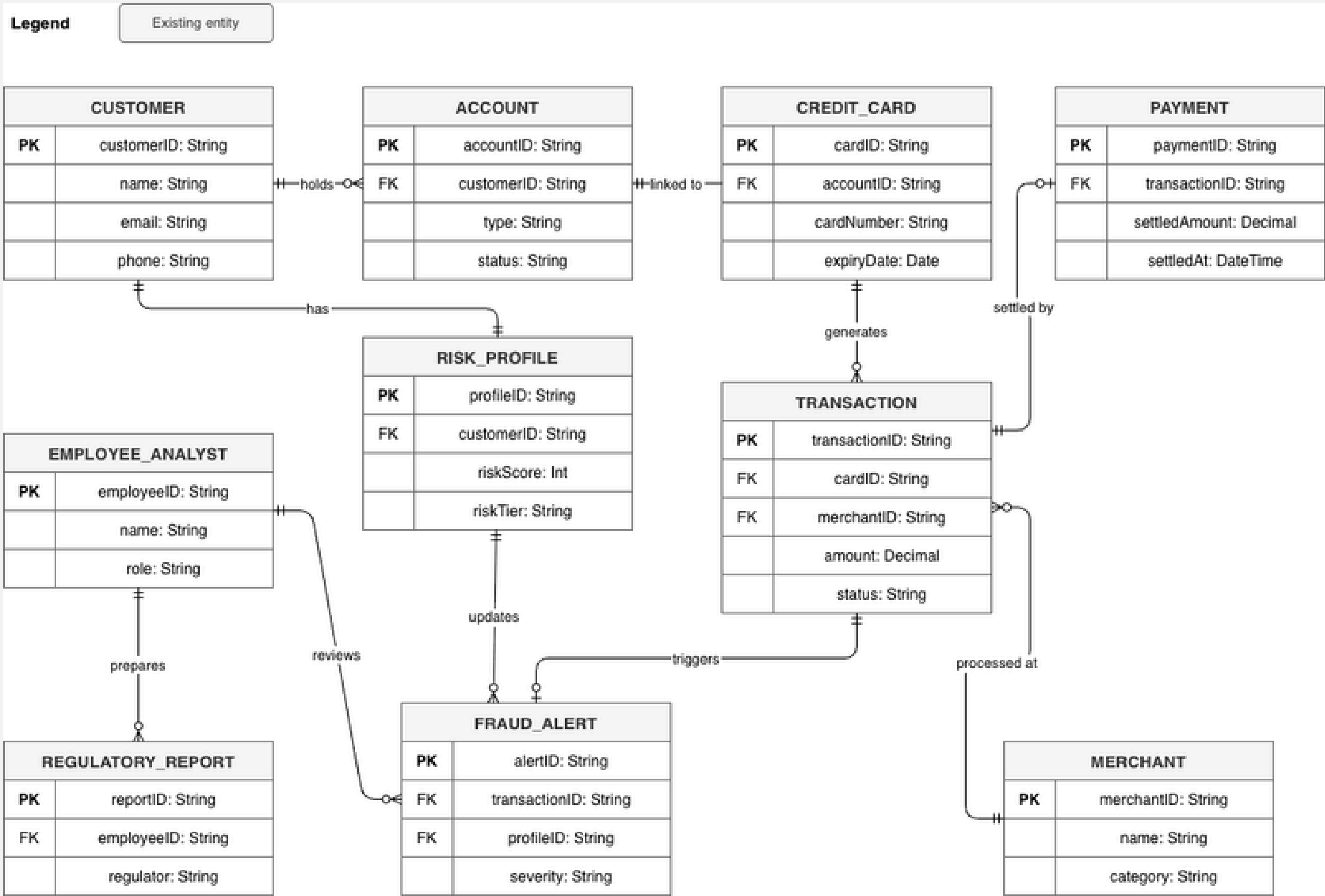




02

Information System Architecture

2.1 The Baseline: What It Can't Do

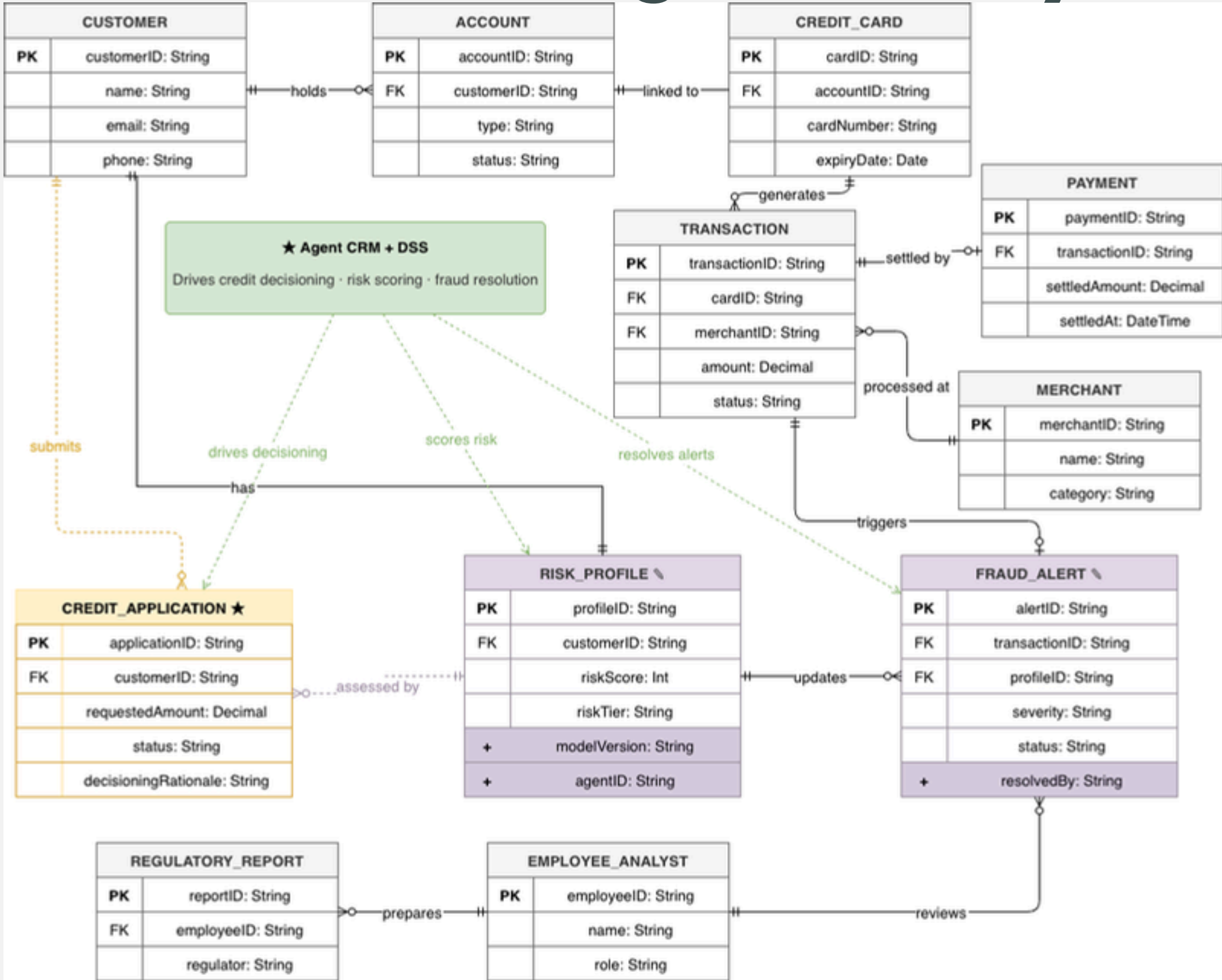


10 entities:

- Customer
- Account
- Credit Card
- Transaction
- Payment
- Merchant
- Risk Profile
- Fraud Alert
- Employee/Analyst
- Regulatory Report

Key Insight:
The baseline captures transactions and fraud, but has no way to act on them.

2.2 What Changed & Why It Matters



Modified

Added resolvedBy. Agents handle low-severity alerts automatically, freeing analysts for high-priority cases.

Modified

Added modelVersion + agentID. Every AI-driven score is traceable and audit-ready.

New Entity

New entity that logs what the agent decided and why — keeping decisions accountable and reversible.

New System

Agent CRM + DSS Pulls live data from transactions, risk, and fraud to drive real-time decisions. 2–3 year rollout.

2.3 What to Watch

Regulatory Explainability	Cold Start Problem	Data Quality & Consistency
• If the AI model gets updated mid-deployment, older decisions may lose their paper trail. This requires upfront planning around version tracking, testing configurations, and rollback procedures, not just locking versions after the fact • Fix: Assign a version ID to every model deployment, log all configuration changes, and define a rollback protocol before go-live.	• The system needs enough historical data to make good recommendations. Early on, suggestions may be off until it learns the patterns. • Fix: Seed the agent with 12 months of historical data before go-live.	• The new fields introduced across entities need to be filled in correctly and consistently; if they're not, the AI will make decisions based on bad data. • Fix: Enforce field validation rules and establish a data governance framework from day one.



03

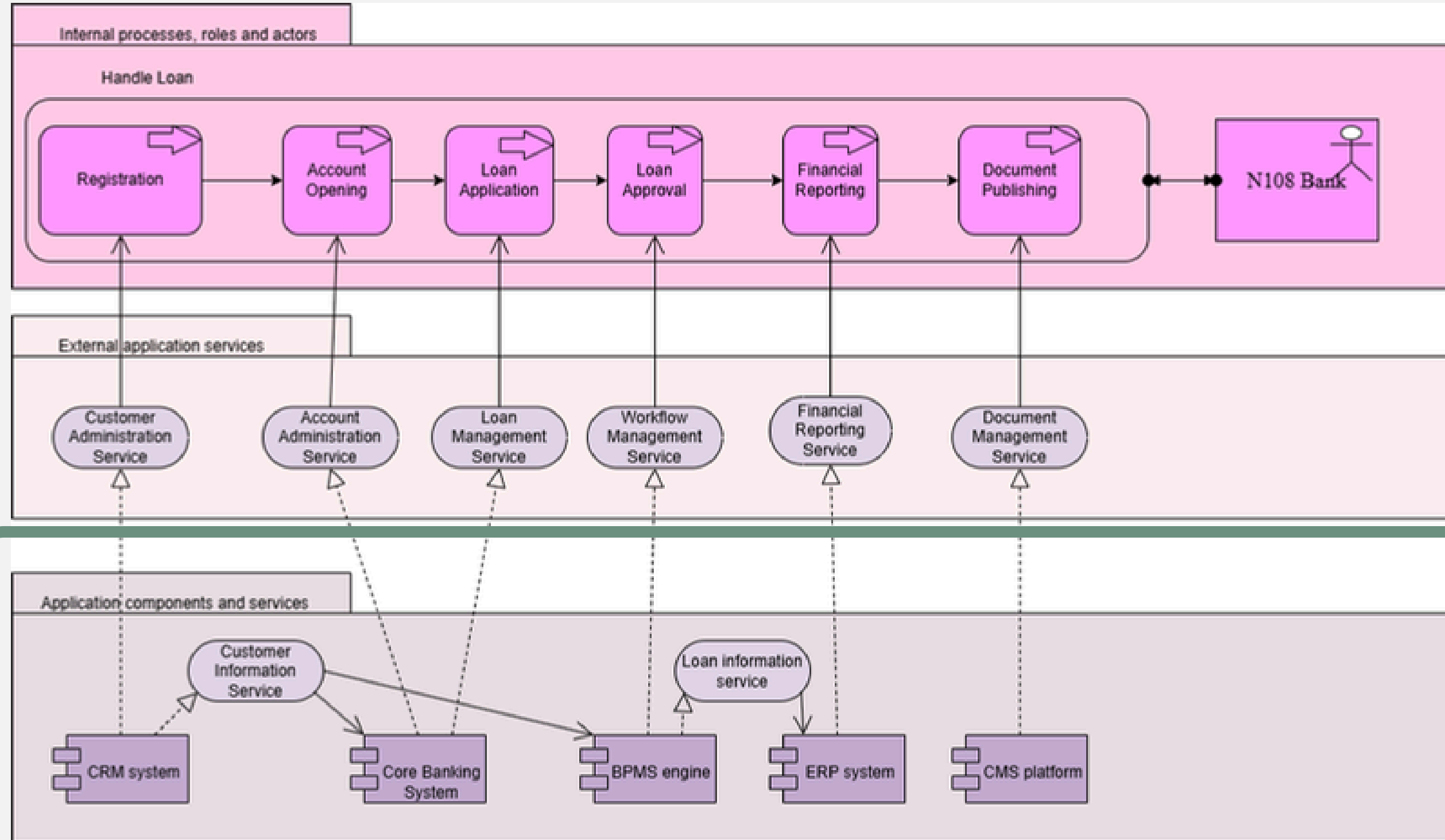
Application System Architecture

3.1.1 Original Layered Architecture

Internal Processes

External Application Services

Application Systems



CRM system

Core Banking system

BPMN engine

ERP system

CMS platform

3.1.2 Application Systems

CRM

- Manage customer data
- Store centralized customer information
- Support manual decision-making

Core Banking System

- Handles account management and loan processing
- Serves as the central transactional platform
- Supports real-time transaction processing

BPMS

- Manages workflows such as loan applications and approvals
- Automates and coordinates key business processes

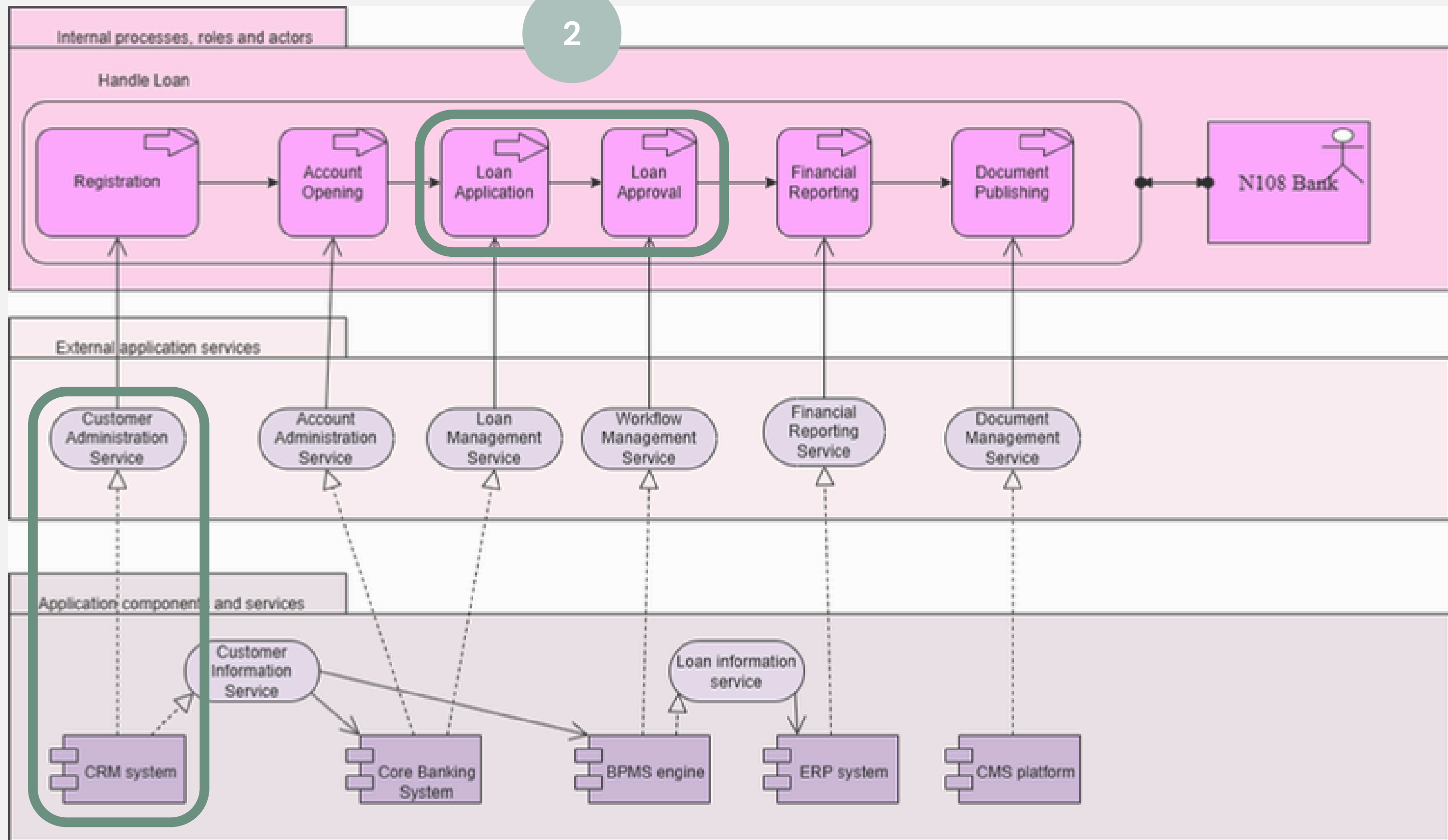
ERP

- Supports internal operations like finance and reporting
- Provides insights for organizational decision-making
- Helps ensure accurate and consistent reporting

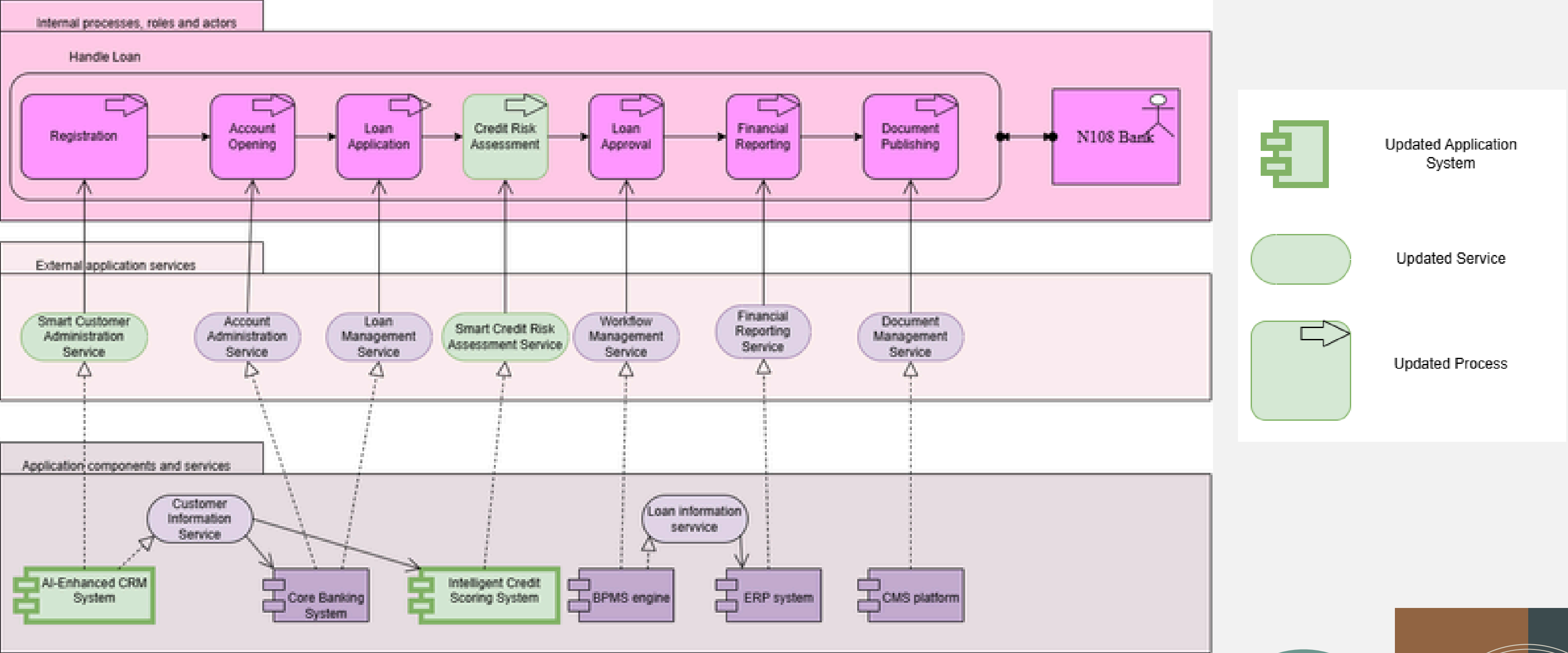
CMS

- Manages digital content and documents
- Ensures consistent communication across channels

(Appian, 2023)
(Buttle & Maklan, 2019)
(Monk & Wagner, 2013)
(Nadkarni & Gupta, 2022)



3.2 Scenario 2: Digital Transformation – Application Layer Changes



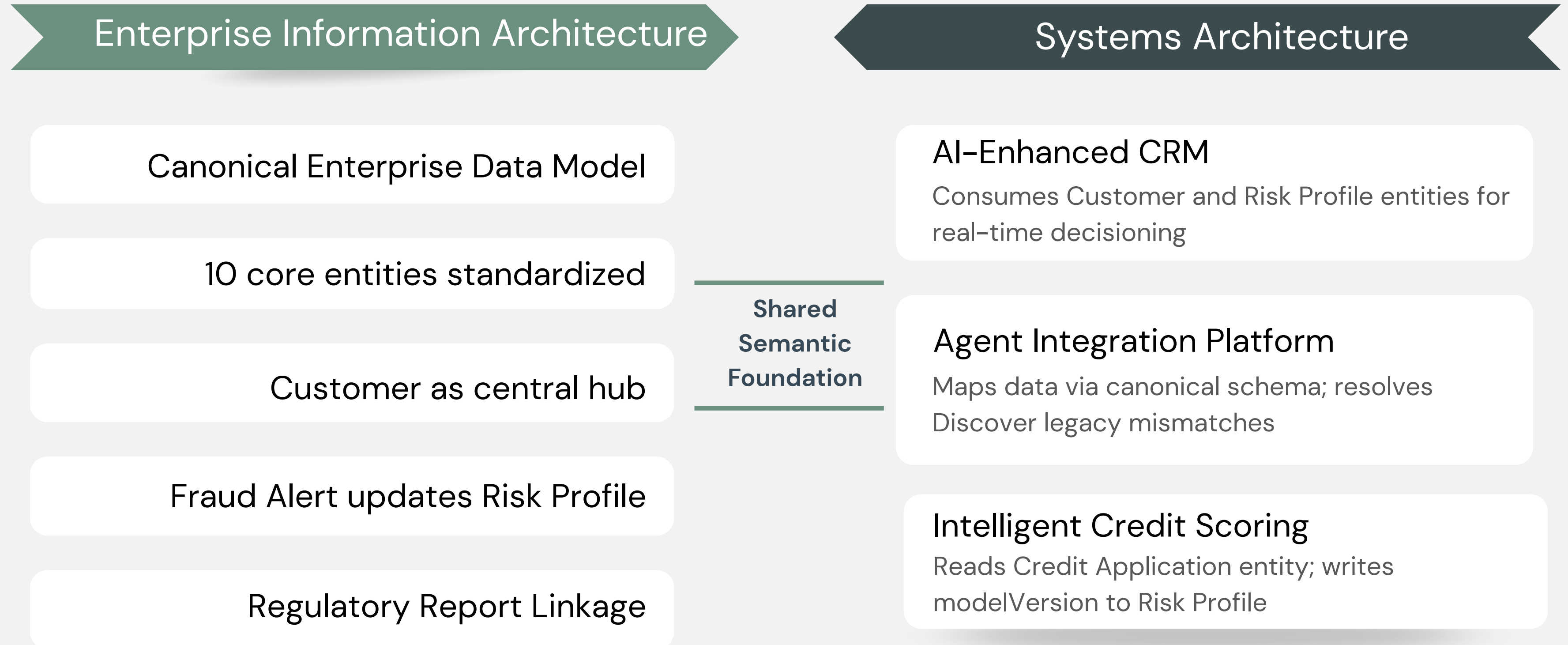
(Salesforce, n.d.)



04

Strategic Alignment

4.1 Canonical EDM: The Alignment Bridge



(Ross, Weill and Robertson, 2006)

(Loshin, 2010)

4.2 Scenario: Digital Transformation

Before: Manual Decision Flows

CRM (data store only)

Manual Credit Evaluation

Human Fraud Review

Delayed Risk Scoring

After: Agentic Systems

AI-Enhanced CRM (UPGRADED)

Real-time personalization and automated customer interactions

Intelligent Credit Scoring (NEW)

Agentic credit risk; writes modelVersion and agentID to Risk Profile

Fraud Alert: ResolvedBy Attribute

Distinguishes human vs agentic resolution for audit trail

Risk Profile: ModelVersion Attribute

Regulatory accountability for all AI-driven decisions

(DAMA International, 2017)

(Lankhorst, Proper and Jonkers, 2018)

4.3 Governance Alignment

Chief Data Officer (NEW)

- Sets enterprise data strategy
- Bridges EIA systems layer
- Resolves federated ownership gaps

Model Risk and AI Governance (NEW)

- Independent model validation (OCC)
- Biannual audits of agentic attributes
- Governs AI-Enhanced CRM and Scoring

Data Owners and Committees

- Integration Event stewardship
- Escalation to Governance Committee
- Compliance: Fed, OCC, CFPB

Cross-Layer Accountability:

- Sets enterprise data strategy
- Bridges EIA systems layer
- Resolves federated ownership gaps

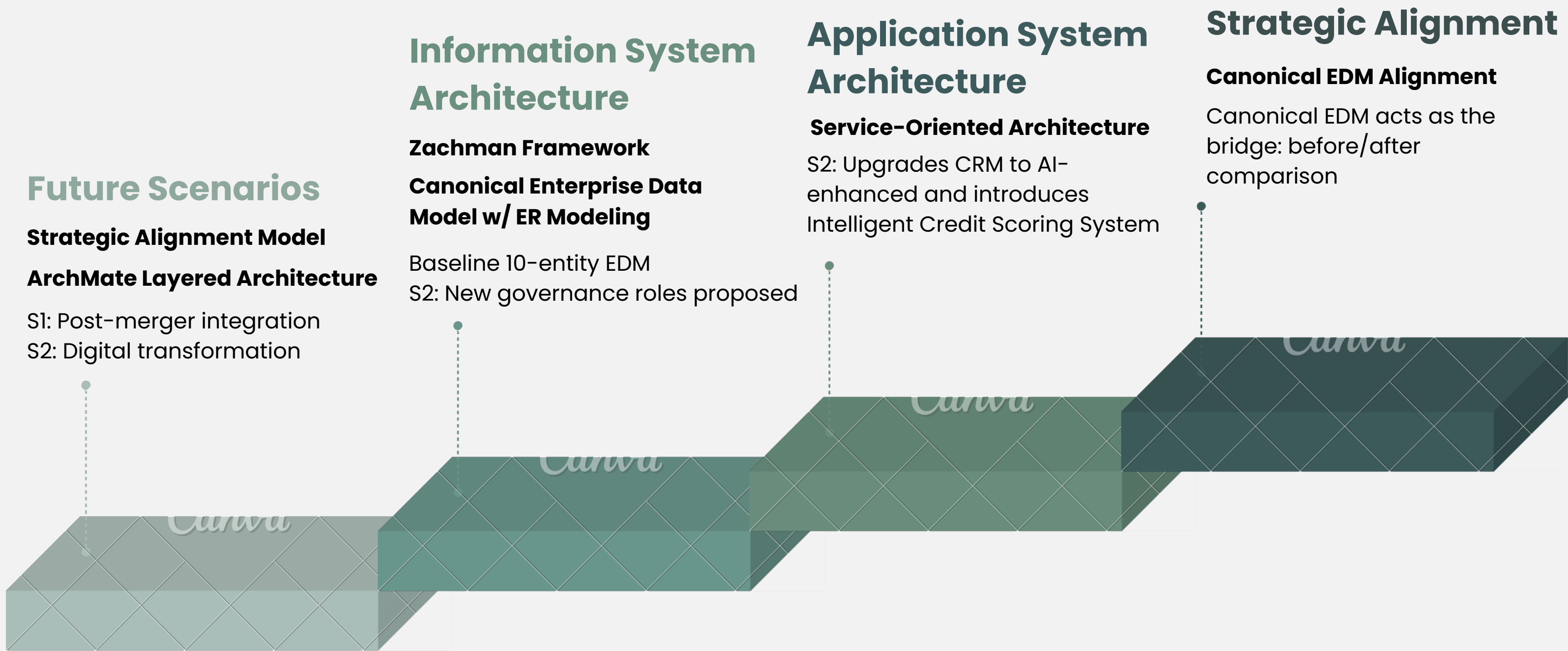
(DAMA International, 2017)



05

Roadmaps

5.1 Analysis Roadmap & Methodology



5.2 Future Opportunites

Predictive Customer Profiling

Predictive customer profiling using longitudinal transaction and interaction history via Agent-enabled CRM

Expanded Payment Network Analytics

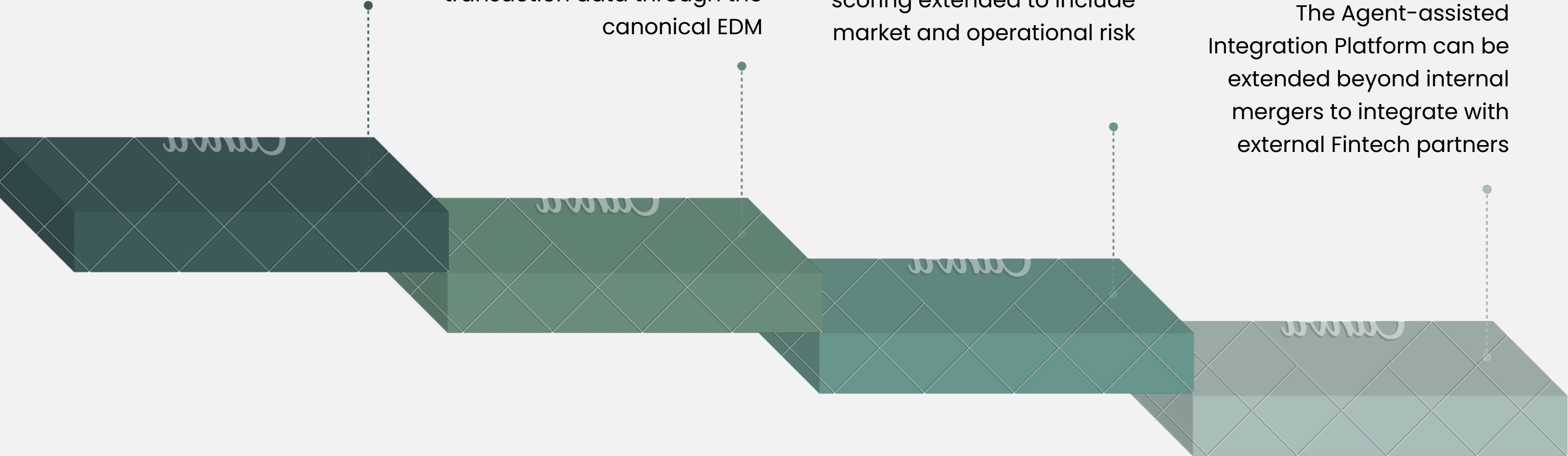
Expanded payment network analytics leveraging Discover transaction data through the canonical EDM

Broader Risk Coverage

Continuous Risk Profile scoring extended to include market and operational risk

Expansion into Ecosystem Banking

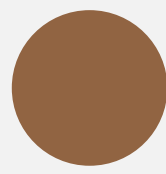
The Agent-assisted Integration Platform can be extended beyond internal mergers to integrate with external Fintech partners





Thank You





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